THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 9: **Cell: The Building Block of Life** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The cell was first observed and named by: 1

(a) Robert Hooke **(b)** Robert Brown
(c) Rudolf Virchow **(d)** Anton van Leeuwenhoek

2. A cell without a well-defined nucleus is a: 1

(a) eukaryotic cell **(b)** animal cell **(c)** prokaryotic cell **(d)** plant cell

3. The endoplasmic reticulum with ribosomes attached is the: 1

(a) vacuole **(b)** smooth ER **(c)** rough ER **(d)** Golgi

4. Cell theory was proposed by: 1

(a) Darwin and Wallace **(b)** Watson and Crick
(c) Schleiden and Schwann **(d)** Hooke and Brown

5. Which of the following is a prokaryote? 1

(a) bacterium **(b)** onion cell **(c)** amoeba **(d)** yeast

6. The smooth ER is concerned with: 1

(a) lipid synthesis and detoxification **(b)** protein synthesis
(c) photosynthesis **(d)** respiration

7. Virchow added to the cell theory that: 1

(a) cells are visible to the eye **(b)** all cells have walls
(c) new cells arise from pre-existing cells **(d)** cells are made of atoms

8. The organelle called the 'powerhouse of the cell' is the: 1

(a) mitochondrion **(b)** Golgi apparatus **(c)** chloroplast **(d)** lysosome

9. Chromosomes are made of: 1

(a) only starch **(b)** DNA and proteins **(c)** only fat **(d)** only water

10. Organisms made of a single cell are called: 1

(a) unicellular **(b)** non-living **(c)** multicellular **(d)** acellular

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Who discovered the cell? What did he observe? 2

12. Name three types of plastids and their functions. 2

13. What is plasmolysis? 2

14. State the cell theory. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A raisin is placed in water and another in a strong sugar solution. Predict and explain what happens. 3

16. Explain why red blood cells and nerve cells have different shapes. 3

17. State three functions of vacuoles. 3

OR

Describe an activity to show osmosis using a potato. 3

18. What is meant by a hypotonic, hypertonic and isotonic solution? Give the effect on a cell. 3

OR

Explain how a cell wall protects a plant cell in a hypotonic solution. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the structure of a typical animal cell with a labelled description of the organelles and their functions. 5

OR

Describe the two types of cell division and their significance. 5

20. Describe the structure of a typical plant cell and state the three structures that distinguish it from an animal cell. 5

OR

Explain the role of mitochondria and plastids in a cell, naming the kinds of plastids. 5

END OF QUESTION PAPER — Total: 40 Marks

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